

Lesson 3 Docker Container Usage

1. Help Command

“docker info” is used to display information about the Docker system, including its configuration and current status. It provides a detailed summary of Docker images, the number of containers, the Docker daemon, storage drivers, networks, and other relevant information.

“docker --help” is used to obtain help information for Docker commands. If it is executed, all available Docker commands and their options will be listed for reference.

2. Image Command

1) “docker images” lists all images on the local master.

```
pi@raspberrypi:~ $ docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
hello-world    latest    ee301c921b8a   8 months ago   9.14kB
```

Corresponding Parameters:

REPOSITORY: the repository source of the image.

TAG: the tag of the image.

IMAGE ID: the ID of the image.

CREATED: the creation time of the image.

SIZE: the size of the image.

(The same repository source can have multiple tags representing different versions of the repository source. You can define different images using REPOSITORY: TAG. If a tag version for an image is not defined, Docker will use the “latest” image by default.)

“docker images” also provides some options:

-a: list all local images

-q: display image id only

--digests: display the image's digest information

2) "docker pull" is used to download images from a selected repository.

```
pi@raspberrypi:~ $ docker pull --help
Usage:  docker pull [OPTIONS] NAME[:TAG|@DIGEST]

Download an image from a registry

Aliases:
  docker image pull, docker pull

Options:
  -a, --all-tags          Download all tagged images in the repository
  --disable-content-trust Skip image verification (default true)
  --platform string       Set platform if server is multi-platform capable
  -q, --quiet             Suppress verbose output
```

Usage:

"docker pull" image repository address/image name: version

Demonstration (it cannot be downloaded, for reference only): "docker pull"

hiwonder/ros-foxy:1.0.0

For example, download the "debian" image: Press "Ctrl+Alt+T" to open the command line terminal. Then enter "docker pull debian" and press "Enter".

```
pi@raspberrypi:~ $ docker pull debian
Using default tag: latest
latest: Pulling from library/debian
5665c1f9a9e1: Pull complete
Digest: sha256:b16cef8cbcb20935c0f052e37fc3d38dc92bfec0bcfb894c328547f81e932d67
Status: Downloaded newer image for debian:latest
docker.io/library/debian:latest
```

Enter "docker images" to view the images on the master, and you can see a "debian" image has been added to the list of images.

```
pi@raspberrypi:~ $ docker images
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
debian               latest          60e0ade36b1a   11 days ago    139MB
hello-world         latest          ee301c921b8a   8 months ago   9.14kB
```

3) "docker search" is used to search the image corresponding to the

“DockerHub” repository.

Take researching the “ROS” image for an example. Press “Ctrl+Alt+T” to open the command line terminal. Then enter “docker search ros” and press “Enter”.

```
pi@raspberrypi:~ $ docker search ros
```

NAME	DESCRIPTION	STARS	OFFICIAL
ros	The Robot Operating System (ROS) is an open ...	621	[OK]
osrf/ros	The Robot Operating System (ROS) is an open ...	151	
rostlab/pp1cs	DockerImage for the exercise of Protein Pred...	0	
osrf/ros2	**Experimental** Docker Images for ROS2 deve...	71	
rosaelng/server	RosaelNG node.js Server	0	
rosaelng/cli	Command Line Interface for RosaelNG	0	
rosaelng/server-java	RosaelNG Java Server	0	
osrf/ros_legacy	Legacy docker image archive for Robot Operat...	3	
rostlab/loctree3	https://github.com/Rostlab/LocTree3	0	
rostlab/somena	https://github.com/Rostlab/someNA	0	
rostlab/aquaria	Aquaria automatic build	0	
rostlab/seqvec	Modelling the Language of Life - Deep Learni...	0	
uobflightlabstarling/rosbridge-suite		0	
rostlab/bio_embeddings		0	
osrf/ros2_packaging_aarch64_jammy_39_snapshot		2	
osrf/ros2_packaging_jammy_39_snapshot		0	
osrf/ros2_batch_ci_aarch64_jammy_39_snapshot		0	
osrf/ros2_batch_ci_jammy_39_snapshot		0	
rostlab/ncov_casp_backend		0	
uobflightlabstarling/ros-web-bridge		0	
uobflightlabstarling/ros_debug		0	
uobflightlabstarling/ros_demo		0	
ardupilot/ardupilot-dev-ros	ArduPilot CI image for ROS2 build	0	
osrf/space-ros	Docker images for the Space ROS project http...	1	
osrf/icra2023_ros2_gz_tutorial	Docker image for the ICRA London tutorial on...	2	

“docker search” also provides some options.

```
pi@raspberrypi:~ $ docker search --help
```

Usage: docker search [OPTIONS] TERM

Search Docker Hub for images

Options:

-f, --filter filter	Filter output based on conditions provided
--format string	Pretty-print search using a Go template
--limit int	Max number of search results
--no-trunc	Don't truncate output

-f, --filter filter: Filter output based on conditions provided

--format string: Pretty-print search using a Go template

--limit int: Max number of research results

--no-trunc: Do not truncate output

4) “docker rmi” is used to delete images.

docker rmi -f [image id]: delete a single image

docker rmi -f [image name: tag] [image name: tag]: delete multiple images

`docker rmi -f $(docker images -qa)` : delete all images

3. Container Command

The containers can be created only if there are images. Use “debian” to test. Press “Ctrl+Alt+T” to open the command line terminal. Then enter “docker pull debian” and press “Enter”.

```
pi@raspberrypi:~ $ docker pull debian
Using default tag: latest
latest: Pulling from library/debian
5665c1f9a9e1: Pull complete
Digest: sha256:b16cef8cbcb20935c0f052e37fc3d38dc92bfec0bcfb894c328547f81e932d67
Status: Downloaded newer image for debian:latest
docker.io/library/debian:latest
```

3.1 “docker run” Image Running and Container Starting

Command Parameters: `docker run [OPTIONS] IMAGE [COMMAND] [ARG...]`

Commonly Used Parameters:

`--name="Name"`: assign a name to the container

`-d`: to run the container in the background and print its id

`-i`: to run the container in interactive mode and be used with `-t`

`-t`: to redistribute a terminal for the container and be used with `-i`

`-P`: to map random ports (uppercase)

`-p` : to map appointed ports (lowercase), generally, with four writing methods

`ip:hostPort:containerPort`

`ip::containerPort`

`hostPort:containerPort` (commonly used)

Take the running “debian” image as an example. In the command line terminal, enter “`docker run -it debian /bin/bash`” and press “Enter”.

“pi@raspberrypi” is the master system interface, and “root@9aeb4cc 795e6” is in the interface inside the “docker” container.

```
pi@raspberrypi:~ $ docker run -it debian /bin/bash
root@9aeb4cc795e6:/#
```

Parameters:

-i: interactive mode

-t: terminal

debian : “debian” image

“/bin/bash” is the command shown after the image name, indicating the hope that there is an interactive “Shell”. Therefore, “/bin/bash” is used.

To exit the terminal, enter “exit” in the terminal and press “Enter”.

```
pi@raspberrypi:~ $ docker run -it debian /bin/bash
root@9aeb4cc795e6:/# exit
exit
pi@raspberrypi:~ $
```

3.2 “docker ps” Listing All Running Containers

Command Parameters: docker ps [OPTIONS]

Commonly Used Parameters:

-a: list all running and previously run containers

-l: display the latest created container

-n=? : display multiple (n) recently created containers

-q: Default mode, only display container IDs

Enter “docker ps -a” in the command line terminal and press “Enter”. The running and run containers are displayed. “container id” is the container ID, “image” is the image name used by the container, “created” is the creation time of the container, and “status” is the current status of the container.

```
pi@raspberrypi:~ $ docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS              PORTS          NAMES
9aeb4cc795e6   debian        "/bin/bash"             10 minutes ago Exited (0) 8 minutes ago                  intelligent_ramanujan
e3fd00242b77   hello-world   "/hello"                 About an hour ago Exited (0) About an hour ago                  infallible_dirac
20b65a5ef772   hello-world   "/hello"                 2 hours ago    Exited (0) 2 hours ago                  romantic_pasteur
```

3.3 Exit Container

There are two commands for exiting containers:

1) Enter “exit” directly in the terminal and press “Enter”. This action will stop the container from running and exit it.

```
pi@raspberrypi:~ $ docker run -it debian /bin/bash
root@9aeb4cc795e6:/# exit
exit
pi@raspberrypi:~ $
```

2) Use “ctrl+P+Q”. The container exits directly but keeps running. You can enter the command “docker ps” in the terminal to view the running containers.

```
pi@raspberrypi:~ $ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
56d746b94607	debian	"/bin/bash"	7 minutes ago	Up 7 minutes		optimistic_kirch

3.4 Multiple Terminals Enter Running Container

You can enter a running container from another terminal with two commands:

“docker attach” is used to enter the terminal of the container start command without starting a new process. Exiting from this container may cause the container to stop running.

“docker exec” is used to open a new terminal in the container and start a new process. Exiting from the container will not affect the container’s running status. This method is generally preferred for entering a container.

3.4.1 docker attach

1) Press “Ctrl+Alt+T” to open the command line terminal and enter “docker ps”. It displays the running containers.

```
pi@raspberrypi:~ $ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
004c7b83e76c	debian	"/bin/bash"	9 seconds ago	Up 6 seconds		quirky_mccarthy
cd1df225f0bf	debian	"/bin/bash"	About a minute ago	Up About a minute		nostalgic_black

2) Enter “docker attach 004c” in the terminal (The container ID can be shortened as long as it is a unique identifier for the container.) to get into the

running container.

```
pi@raspberrypi:~ $ docker attach 004c
root@004c7b83e76c:/#
```

3) Enter “exit” in the terminal and press “Enter”. The container stops running and exits.

```
pi@raspberrypi:~ $ docker attach 004c
root@004c7b83e76c:/# exit
exit
```

4) Enter “docker ps” in the terminal. It displays the running container, and you can see the earlier exited container is also closed.

```
pi@raspberrypi:~ $ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
cd1df225f0bf	debian	"/bin/bash"	6 minutes ago	Up 6 minutes		nostalgic_black

3.4.2 docker exec

1) Press “Ctrl+Alt+T” to open the command line terminal and enter “docker ps”. It displays the running containers.

```
pi@raspberrypi:~ $ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
efe969b704a7	debian	"/bin/bash"	8 seconds ago	Up 5 seconds		fervent_williams
cd1df225f0bf	debian	"/bin/bash"	8 minutes ago	Up 7 minutes		nostalgic_black

2) Enter “docker exec -it efe9 /bin/bash” in the terminal (The container ID can be shortened as long as it is a unique identifier for the container.) to enter the running container.

```
pi@raspberrypi:~ $ docker exec -it efe9 /bin/bash
```

3) Enter “exit” in the terminal and press “Enter”. The container will exit.

```
pi@raspberrypi:~ $ docker exec -it efe9 /bin/bash
root@efe969b704a7:/# exit
exit
```


4) Enter “docker ps” in the terminal. It displays the running container, and you can see the exited container is not closed.

```
pi@raspberrypi:~$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS          NAMES
efe969b704a7   debian   "/bin/bash"             3 minutes ago Up 3 minutes           fervent_williams
cd1df225f0bf   debian   "/bin/bash"             11 minutes ago Up 11 minutes          nostalgic_black
```

3.5 Start and Stop Container

docker start [container id or container name]: to start a container

docker restart [container id or container name]: to restart a container

docker stop [container id or container name]: to stop a running container

docker kill [container id or container name]: to forcefully stop a running container

3.6 Delete Container

docker rm [container id]: Delete specified container

docker rm -f \$(docker ps -a -q): Delete all containers

docker ps -a -q|xargs docker rm: Delete all containers

4. Other Commonly-Used Other Commands

4.1 View Process Information Running in Container

1) Press “Ctrl+Alt+T” to open the command line terminal and enter “docker ps”. It displays the running containers.

```
pi@raspberrypi:~$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS          NAMES
efe969b704a7   debian   "/bin/bash"             8 seconds ago Up 5 seconds           fervent_williams
cd1df225f0bf   debian   "/bin/bash"             8 minutes ago Up 7 minutes          nostalgic_black
```

2) Enter “docker top efe9” in the terminal. It displays the process information running in the container.

```
pi@raspberrypi:~$ docker top efe9
UID          PID          PPID         C
STIME        TTY          TIME         CMD
root         3186         3163         0
18:01        pts/0        00:00:00     /bin/bash
```


Corresponding Parameters:

UID: User ID.

PID: Process ID.

PPID: Parent Process ID.

C: CPU utilization of the process.

STIME: Start time of the process.

TTY: Terminal associated with the process.

TIME: Total CPU time used by the process.

CMD: Command line used to start the process.

4.2 View Metadata of Container or Image

1) Press “Ctrl+Alt+T” to open the command line terminal and enter “docker ps”. It displays the running containers.

```
pi@raspberrypi:~ $ docker ps
CONTAINER ID   IMAGE    COMMAND                  CREATED        STATUS        PORTS        NAMES
efe969b704a7   debian   "/bin/bash"             8 seconds ago Up 5 seconds          fervent_williams
cd1df225f0bf   debian   "/bin/bash"             8 minutes ago Up 7 minutes          nostalgic_black
```

2) Enter “docker inspect efe9” in the terminal to output the JSON data including detailed information about the container or image.

```
pi@raspberrypi:~ $ docker inspect efe9
[
  {
    "Id": "efe969b704a7bbbc02a8ae84c05993e89f70fe86b9a6e53df5933c99f0a0d637",
    "Created": "2024-01-22T10:01:09.554324422Z",
    "Path": "/bin/bash",
    "Args": [],
    "State": {
      "Status": "running",
      "Running": true,
      "Paused": false,
      "Restarting": false,
      "OOMKilled": false,
      "Dead": false,
      "Pid": 3186,
      "ExitCode": 0,
      "Error": "",
      "StartedAt": "2024-01-22T10:01:11.688664835Z",
      "FinishedAt": "0001-01-01T00:00:00Z"
    },
    "Image": "sha256:60e0ade36b1ab9227b0b8fc5a9731a3a5b9217e9feeb4ba7a2f8cb84addef87b",
    "ResolvConfPath": "/var/lib/docker/containers/efe969b704a7bbbc02a8ae84c05993e89f70fe86b9a6e53df5933c99f0a0d637/resolv.conf",
    "HostnamePath": "/var/lib/docker/containers/efe969b704a7bbbc02a8ae84c05993e89f70fe86b9a6e53df5933c99f0a0d637/hostname",
    "HostsPath": "/var/lib/docker/containers/efe969b704a7bbbc02a8ae84c05993e89f70fe86b9a6e53df5933c99f0a0d637/hosts",
    "LogPath": "/var/lib/docker/containers/efe969b704a7bbbc02a8ae84c05993e89f70fe86b9a6e53df5933c99f0a0d637/efe969b704a7bbbc02a8ae84c05993e89f70fe86b9a6e53df5933c99f0a0d637-json.log",
    "Name": "/fervent_williams",
    "RestartCount": 0
  }
]
```